

Pre-Research Memo: A Benchmark and Agentic Baseline for AI Paper Reproducibility

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Project in one sentence

Build two co-equal artifacts on DeltaAI: a stratified benchmark of recent AI papers (*ReproBench*) and an agentic CLI (*reprocli*) that attempts non-interactive end-to-end reproduction, together producing the first measured curve of how reproduction success scales with agent compute budget across the current AI literature.

Motivation

A growing fraction of AI research ships with code, weights, and data, yet reproducing a result from a paper still demands hours to weeks of skilled human work. Measuring where on this spectrum an agent succeeds, and how that success rate scales with the compute we give it, yields (a) a capability metric that tracks frontier models over time, (b) the first quantitative read on the marginal cost of automated reproducibility, and (c) a tool that removes meaningful labor from the research workflow even at modest success rates. The compute-versus-score curve is the central empirical contribution that ties the two artifacts together.

Benchmark Design: *ReproBench*

Source corpus. NeurIPS 2025 final camera-ready papers and supplementary material, with roughly six months for repositories and checkpoint mirrors to stabilise. Target: 250 to 500 papers after filtering for in-scope compute.

Minimum reproducible claim. For each paper we identify the smallest experimental setting reported in the paper that still constitutes a non-trivial result, and lock that as the reproduction target. Concretely, if a paper releases JiT-S, JiT-B, and JiT-L, we target JiT-S. If a paper sweeps five datasets, we target the cheapest. The headline number for that minimum setting, lifted from a table or plot with an explicit tolerance, becomes the ground-truth claim. This keeps the benchmark tractable on DeltaAI while preserving the full end-to-end pipeline (data, code, training, evaluation) the agent has to navigate.

Stratification axes. Each paper is labelled along four artifact-availability dimensions, then bucketed into a difficulty tier:

- *Code*: official repository present, partial, or absent.
- *Model weights* for the minimum claim: released, partial (e.g. tokenizer only), or absent.
- *Training/eval data*: open, gated, or proprietary.
- *Compute footprint* of the minimum claim: fits a single GH200 node, multi-node, or out of scope.

Difficulty tiers.

1. **Tier 1 (warm start)**: code + weights + open data, single-node eval-only reproduction.
2. **Tier 2 (light training)**: code + open data, weights need to be retrained or fine-tuned at the minimum scale.
3. **Tier 3 (code only)**: official repository exists but no weights; agent must source data and train.
4. **Tier 4 (paper only)**: no official code; agent reconstructs the implementation from the paper text.
5. **Tier 5 (out of scope)**: compute beyond DeltaAI or proprietary data. Logged but not evaluated, kept for future-proofing the benchmark as compute grows.

Scoring. Pass/fail against the minimum claim within tolerance, with secondary metrics for wall-clock time, GPU-hours consumed, and number of agent retries.

Agent: *reprocli*

Interface. A single command: `reprocli replicate <arxiv-url>`. Non-interactive by design so it can be batch-submitted to SLURM and left to run. Intermediate state is checkpointed to disk so jobs can be queued after preemption.

Backbone. A frontier open-weight reasoning model (target: DeepSeek V4, with Qwen3 and Llama 4 as

fallbacks) served via vLLM on a dedicated GH200 using the [vLLM recipes](#) as the serving template. Agent reasoning and the workloads it spawns run on separate SLURM allocations so long training jobs do not block inference.

Tool surface.

- File system, shell, and Python execution inside a per-task sandboxed workspace.
- arXiv and OpenReview fetchers, GitHub clone, and a paper-to-repository linker.
- `sbatch/squeue` wrapper for spawning training and evaluation jobs on DeltaAI.
- A claim parser that extracts the minimum-claim target metric and a results-checker that compares produced numbers against it within tolerance.

Loop. A ReAct-style scaffold (plan, act, observe, revise) with explicit budgets on tokens, wall-clock, and GPU-hours per paper. The agent terminates with `REPRODUCED`, `FAILED`, or `OVER_BUDGET`, and every run emits a structured trace for later analysis.

Compute budget as a first-class variable. Each paper is run at multiple per-paper compute budgets, with a default sweep of 2, 8, 32, and 128 GH200-hours. The output is a set of *compute-versus-score* curves, one per difficulty tier, that quantify how much compute is required to recover what fraction of the NeurIPS 2025 literature. The shape of these curves (linear, log, saturating) is the headline empirical result.

Rough 10-Week Timeline

- **W1 (Jun 1):** vLLM serving and agent backbone stood up on a GH200; minimum-claim extraction methodology drafted.
- **W2:** NeurIPS 2025 corpus pull, automated metadata extraction, first-pass tier labels.
- **W3:** Manual audit of tier labels and minimum-claim selection on a 50-paper pilot; schema frozen.
- **W4:** *reprocli* skeleton (CLI, SLURM wrapper, tool surface, checkpointing) with end-to-end smoke test on one Tier 1 paper.
- **W5:** Agent loop and claim-verifier integrated; baseline run on the full Tier 1 pilot at a fixed budget.
- **W6:** First baseline at scale on Tier 1 and Tier 2 (target: 100 papers); failure-mode triage.
- **W7:** Extend evaluation to Tier 3 and Tier 4; ablate the highest-cost tools.
- **W8:** Scale to the full 250 to 500 paper benchmark at a fixed budget; corpus finalised.
- **W9:** Compute-budget sweep (2, 8, 32, 128 GH200-hours per paper); first compute-versus-score curves; model ablations (DeepSeek V4 vs. Qwen3 vs. Llama 4).
- **W10:** Writeup, open-source release of benchmark and CLI, internal NCSA presentation.

Open Questions for Our Meeting

1. **Compute envelope:** proposed sweep of 2, 8, 32, 128 GH200-hours per paper. Is a higher upper tier (e.g. 512) worth budgeting for, or does it cannibalise too much DeltaAI time?
2. **Minimum-claim selection:** hand-curated for all 250 to 500 papers, LLM-extracted with human audit on a sample, or a hybrid (LLM proposes, human ratifies)?
3. **Human-in-the-loop policy:** fully autonomous track only, or also a parallel "one hint per paper" track that separates agent capability from benchmark difficulty?
4. **Trace release:** how detailed should the public release of agent traces be (full traces, redacted summaries, or only aggregate metrics)?